

Compressive Single-Photon 3D Cameras

Research Report: Coding Schemes and Pulse-Width Simulations

By Aashish Mankala

1. Introduction

Single-photon avalanche diodes (SPADs) represent a transformative advancement in imaging technology. Unlike digital cameras that have conventional pixels that integrate light intensity over micro- to millisecond timescales, a SPAD pixel can register the arrival of a single photon and track its exact arrival time with sub-nanosecond precision. This unique sensitivity represents a significant step forward for precise 3D imaging, particularly where timing resolution is crucial.

The ability to detect photon time-of-arrival allows SPAD-based systems to estimate depth using the principle of time-of-flight (ToF), where the distance is inferred from how long light takes to travel to an object and back. As a result, SPAD imaging has become increasingly relevant for applications such as robotics, autonomous navigation, medical imaging, fluorescence lifetime imaging microscopy (FLIM), non-line-of-sight (NLOS) imaging, and emerging LiDAR systems. These applications benefit directly from the ability of SPAD sensors to operate under extremely low-light conditions while maintaining high depth precision.

Despite these advantages, SPAD-based 3D cameras face a fundamental challenge, which is data bandwidth. Each detected photon produces a timestamp, meaning the sensor records the photon's time-of-arrival relative to the emitted light pulse. SPAD systems often reduce this raw stream of timestamps by building a time-of-flight histogram at each pixel. The time axis is divided into many small intervals called time bins, and each photon increments the count of the bin corresponding to its arrival time. The time-bin width determines the temporal (and therefore depth) resolution—smaller bins mean finer depth precision. However, histogramming can still produce enormous data volumes. A typical high-resolution histogram may require thousands of time bins per pixel, and when this is repeated across large-format (near-megapixel) SPAD arrays at video frame rates, the total output can still reach tens of gigabytes per second. As the number of pixels, temporal bins, and frame rate increase, the bandwidth required to transmit full per-pixel histograms becomes impractical for standard data transfer interfaces such as USB.

This challenge proposes a trade-off in SPAD-based 3D imaging: achieving high depth resolution requires an increased number of temporal bins, but this increases data bandwidth requirements. Several strategies have been explored to mitigate this issue. Coarse histogramming is when you merge many fine time bins into fewer, wider bins when building the time-of-flight histogram. This reduces data size because you store fewer numbers per pixel, but it blurs photon arrival times, so the depth estimate becomes less precise. Coarse histogramming reduces the number of bins but sacrifices depth precision, while direct timestamp transfer preserves timing accuracy but still produces prohibitively large data streams. As a result, neither approach offers a scalable solution for high-resolution, real-time SPAD imaging.

The central problem, therefore, is how SPAD-based 3D cameras can preserve high-quality depth information while significantly reducing the amount of data that must be processed and transmitted. Addressing this challenge is essential for making single-photon 3D imaging practical beyond laboratory settings.

To solve this problem, Gutierrez-Barragan, Velten, and Gupta et al. proposed Compressive Single-Photon Histograms (CSPH), a framework that compresses photon timing information at the sensor level using carefully designed coding matrices. Rather than constructing full temporal histograms, CSPH projects photon arrivals onto a smaller set of compressive measurements, drastically reducing bandwidth while retaining sufficient information for accurate depth reconstruction.

In this report, I replicate and extend the simulation results presented in the CVPR 2022 paper Compressive Single-Photon 3D Cameras. Using the authors' released codebase, I evaluate multiple coding schemes across different scenes and noise conditions, and I further investigate the effect of varying pulse width on depth reconstruction performance.

2. Background

Single-photon avalanche diode (SPAD) sensors enable 3D imaging by measuring the time-of-flight (ToF) of light at extremely high temporal resolution. When a photon returns after reflecting off an object, the SPAD pixel registers its arrival time with sub-nanosecond precision. Traditional SPAD systems construct a transient histogram, where each bin corresponds to a time interval.

2.1. Transient Histograms and Depth Estimation

A conventional SPAD-based 3D camera builds a full histogram for every pixel, often containing thousands of time bins. The scene depth is computed by identifying the peak in the histogram, which corresponds to the most likely photon time-of-arrival. The temporal resolution of the histogram determines the depth precision, and finer binning produces more accurate depth estimates. However, the number of histogram bins directly determines the data bandwidth. A megapixel SPAD camera with thousands of bins per pixel produces data at a rate that grows linearly with both pixel count and temporal resolution.

2.2. The Bandwidth Problem in SPAD Imaging

As mentioned previously, SPAD-based 3D imaging achieves high depth resolution but at a cost of an increase in data volume. Because each pixel stores a histogram over many time bins, increasing the number of bins directly increases the total data output, quickly resulting in impractically high data rates. Also, as resolution and frame rate increase, the total bandwidth grows exponentially.

Conventional strategies to reduce data bandwidth have significant drawbacks. Coarse histogramming collapses many fine bins into fewer broad bins, reducing data size but sacrificing temporal precision. Timestamp transfer outputs individual photon arrival times directly, but still produces very large data streams. Neither approach maintains both high accuracy and bandwidth efficiency, motivating the need for new compression methods.

2.3. Compressive Single-Photon Histograms (CSPH)

The CVPR 2022 paper introduced Compressive Single-Photon Histograms (CSPH) as a solution. Instead of recording full histograms, CSPH projects the incoming photon stream onto a much smaller number of compressive measurements using a coding matrix C . Each photon contributes to a weighted sum of several compressive bins, which allows the system to retain its structure while reducing the output data. The compression ratio N/K can reach 50–100 $\times$ while preserving usable depth accuracy, where N is the number of fine-grained temporal bins, and K is the number of compressive bins.

2.4. Single-Photon Histogram Formation in SPAD Sensors

SPAD-based 3D cameras estimate depth by measuring the time-of-arrival of photons reflected from a scene after a pulsed illumination event. For each laser pulse, a SPAD pixel becomes active and records the arrival time of the first detected photon before entering a short dead time. This process is repeated over many laser cycles, allowing the sensor to accumulate photon arrival information across time.

Rather than storing individual timestamps, the recorded photon arrivals are sent into a per-pixel histogram. The time axis is divided into discrete time bins, and each detected photon increments the corresponding bin. Over many repetitions, this histogram approximates the underlying photon flux waveform, whose peak location corresponds to the photon time-of-flight and thus the scene depth. While histogramming reduces raw timestamp data, the need for thousands of time bins per pixel still leads to extremely high data rates for large SPAD arrays, motivating the use of compressive strategies such as CSPH.

2.5. Properties of a Good Coding Matrix

The effectiveness of CSPH depends heavily on the design of the coding matrix. The paper establishes four key properties. The first is uniqueness, which explains that no two time bins should map to the same code. Otherwise, different depths become indistinguishable. The second is locality & robustness, which says nearby time bins should map to similar codes. This ensures stable performance even under noise and low-photon conditions. The third is multipath resistance, where the coding should be insensitive to indirect reflections, which otherwise distort transient histograms. The final is band-limit compatibility, which requires the coding matrix to match the bandwidth of the system's impulse response. Codes with energy outside the sensor's temporal bandwidth produce no useful information. These four properties guide the selection of coding schemes and pulse widths during simulation.

2.6. Coding Schemes Evaluated in the Paper

The authors evaluate several coding matrices, each offering distinct advantages depending on the desired compression level and noise conditions. Truncated Fourier coding uses low-frequency Fourier components, making it effective for moderate compression ratios and relatively robust to multipath interference. Gray coding, inspired by continuous analogs of binary Gray codes, is highly locality-preserving and therefore well-suited for aggressive compression where only a small number of compressive bins are used. P-Series coding, which includes both Fourier-based and Gray-based variants, provides a complementary approach that blends the structure of Fourier codes with the stability and locality of Gray codes, often achieving a strong balance between detail preservation and noise robustness. Gated coding

relies on sequential temporal windows as coarse approximations of the transient signal, offering a simple and interpretable compression model. Finally, Identity coding, which corresponds to transmitting full histograms without compression, serves as the baseline for evaluating the performance of compressed schemes. These are the same coding strategies explored in my replication experiments.

2.7. Depth Reconstruction via ZNCC

Once the compressive measurements are collected, CSPH reconstructs depth using a zero-mean normalized cross-correlation (ZNCC) between the observed compressive vector and precomputed templates corresponding to each possible depth. The depth corresponding to the highest correlation score is selected as the estimated time-of-arrival.

3. Experiment Setup

To investigate how different coding schemes and pulse-width settings affect depth reconstruction quality, I performed controlled simulations using the official codebase from Compressive Single-Photon 3D Cameras. All experiments were run on two indoor scenes, Kitchen-2 and Bathroom-Cycles-2, which include rendered transient histograms (2000 time bins), RGB images, and ground-truth depth maps. These pre-rendered datasets allow for controlled testing of noise, compression, and coding design without requiring physical SPAD hardware. The primary script used for evaluation, `eval_coding_flash_lidar_scene.py`, simulates photon detection, applies a chosen coding scheme, reconstructs depth via ZNCC, and outputs both depth images and error maps. Across all experiments, several parameters were held constant to ensure comparability, where the two variables tested were different coding schemes and pulse width values.

4. Coding Schemes Results & Analysis

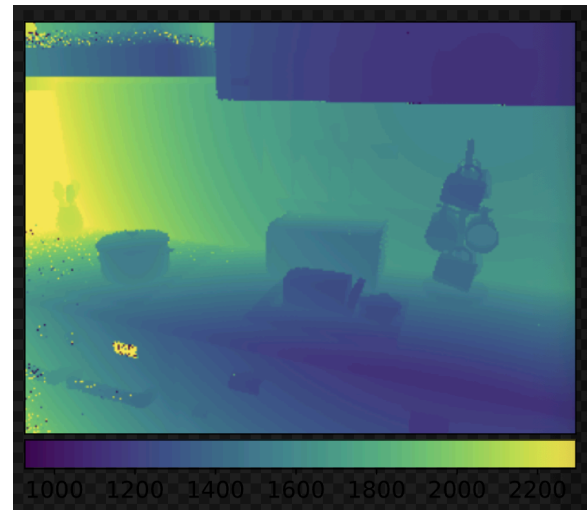
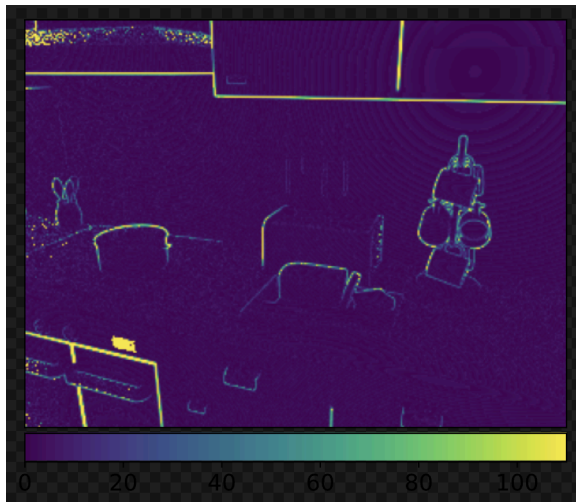
To evaluate the behavior of different coding strategies under varying signal and scene conditions, I applied each coding scheme to two flash LiDAR scenes: Kitchen-2 and Bathroom-Cycles-2. Both scenes contain surfaces with diverse reflective properties, complex geometry, and varying depths. This makes them ideal test cases for replicating the types of evaluations performed in the CSPH paper.

Each result includes mean absolute error (MAE), median absolute error, and percentile error statistics. Along with this, an RGB image, ground-truth depth, reconstructed depth map, and absolute error map are provided, allowing visual comparison between methods.

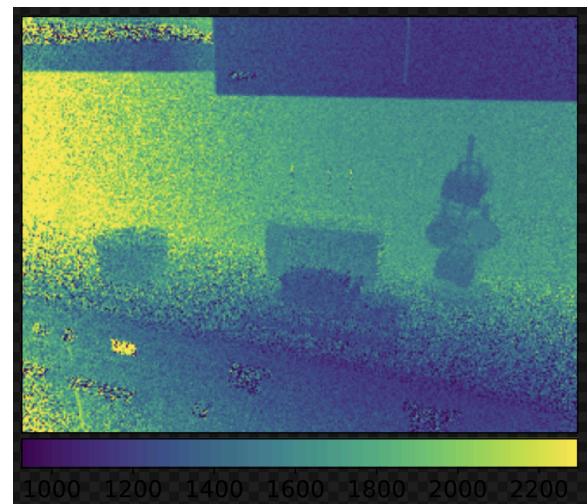
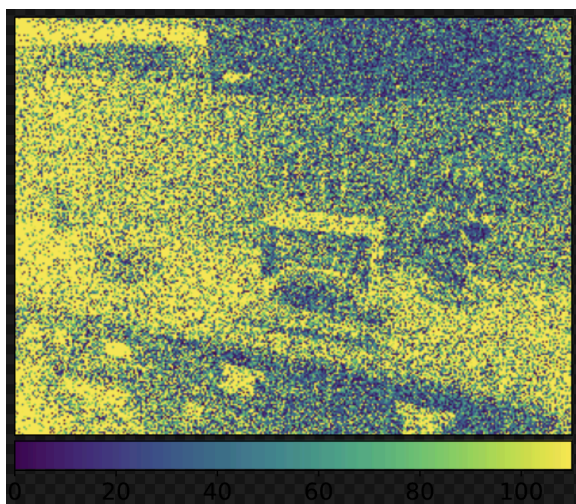
4.1. Results on the Kitchen Scene

1. **Fourier Coding:** The Fourier coding reconstruction for the kitchen scene (Figure F1.b) produced depth estimates that were noticeably noisy and spatially unstable. Large regions of the scene exhibited different depth values, and structural details were lost. This instability was reflected in the high MAE of 117.14 mm. The error patterns (Figure F1.a) displayed speckle-like artifacts across broad surfaces, indicating that Fourier coding struggled to maintain locality and robustness under the deep compression used in the simulation. The result closely resembles the failures described in the original

paper, where pure Fourier coding was shown to be highly sensitive to noise and insufficient for accurate reconstruction in low-photon or high-SBR settings.



2. **Gray Coding:** Gray coding performed better than Fourier in the Kitchen scene. The resulting depth image (Figure G1.b) was far smoother, with well-preserved edges and object boundaries. Most planar surfaces were reconstructed with high stability, and only minor regions around shadow boundaries or highly reflective surfaces showed ambiguity. The relatively low MAE of 17.48 mm demonstrated that Gray coding's locality-preserving structure improves robustness in compressed measurements. The error map (Figure G1.a) confirmed that most of the scene maintained low depth error, with only narrow bands of larger deviation. The reconstruction quality aligns closely with the paper's characterization of Gray coding as one of the most reliable strategies for deep compression.



3. **PSeries Fourier Coding:** The PSeries Fourier scheme produced one of the strongest reconstructions in the Kitchen scene. The depth output (Figure PSF1.b) was smooth,

detailed, and highly consistent with the ground truth. Flat surfaces were cleanly estimated without the instabilities seen in Fourier coding, and the overall MAE of 12.03 mm was significantly lower than all coding schemes except the full histogram reference. PSeries Fourier provided a clear improvement over both Gray and Truncated Fourier, confirming its role as an effective complement method that balances information with stability (Figure PSF1.a). These results reflect the findings in the CSPH paper, which emphasized that PSeries complementary variants often outperform both Gray and standard Fourier when photons are limited.

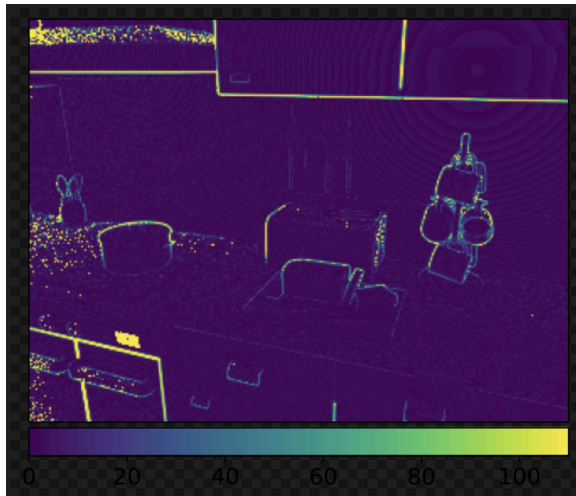


Figure PSF1.a

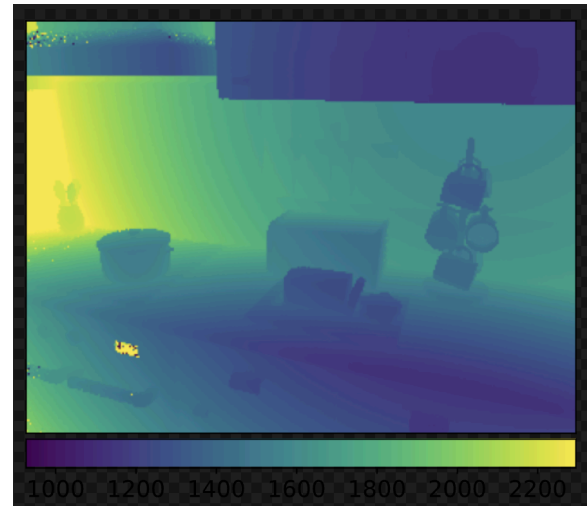


Figure PSF1.b

4. **Truncated Fourier Coding:** Truncated Fourier coding improved noticeably upon the full Fourier method but still exhibited significant spatial noise. The resulting MAE of 22.37 mm falls in a mid-range band, confirming that truncating high-frequency Fourier components reduces some sensitivity to noise but does not fully solve the locality and robustness issues inherent to Fourier-based coding. The reconstructed depth (Figure TF1.b) still contained noise but exhibited far fewer catastrophic failures. Flat surfaces appeared smoother, though edges remained unstable. The error map (Figure TF1.a) showed medium-level noise, consistent with partial robustness due to removing high-frequency Fourier terms.

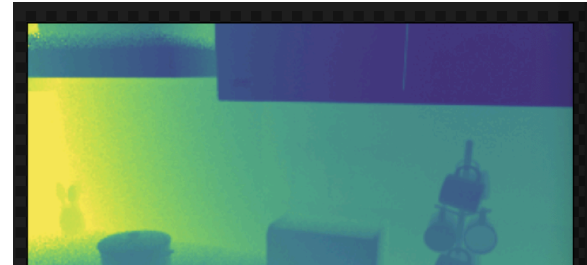


Figure TF1.a

Figure TF1.b

5. **PSeries Gray Coding:** PSeries Gray coding produced depth reconstructions (Figure PSG1.b) that were the closest to the Kitchen scene. The depth map exhibits excellent structure where surfaces are smooth and continuous, edges are crisp, and very few obvious objects are visible. The corresponding depth error map (Figure PSG1.a) shows low error across most of the image, with only small isolated regions of elevated error in challenging areas. The MAE of 11.18 mm was one of the lowest among all compressed coding strategies, rivaling the performance of PSeries Fourier and approaching that of the full histogram baseline. These results strongly support the CSPH paper's conclusion that Gray-based codes offer an optimal balance between locality, robustness, and information richness under compression.

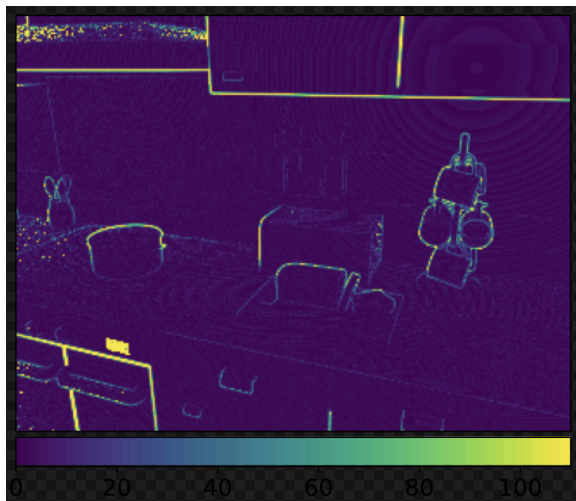


Figure PSG1.a

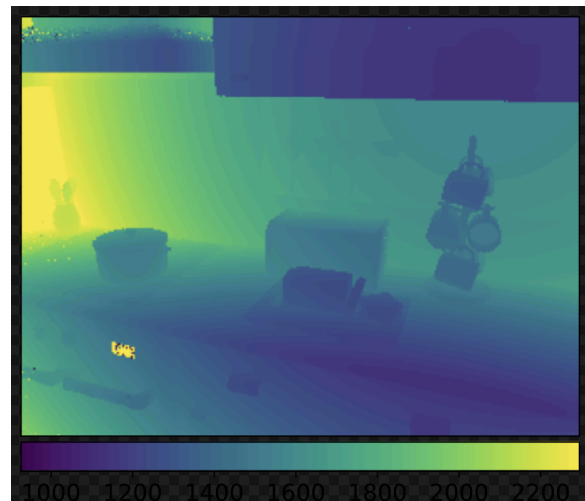


Figure PSG1.b

6. **Gated Coding:** Gated coding produced depth reconstructions (Figure GA1.b) that were visibly coarse and quantized. Continuous surfaces are approximated by step-like bands, and depth transitions appear abrupt rather than smooth. The absolute depth error (Figure GA1.a) map shows broad regions with moderate to high error, especially near

depth boundaries and fine geometric features. The MAE of 40.41 mm is noticeably higher than for Gray and PSeries-based schemes, reflecting the limited effective resolution imposed by the relatively small number of temporal gates. This behavior is consistent with the original paper's assessment. While gating is conceptually simple and hardware-friendly, it sacrifices depth precision and is not competitive with more sophisticated coding strategies at the photon levels used in this experiment.

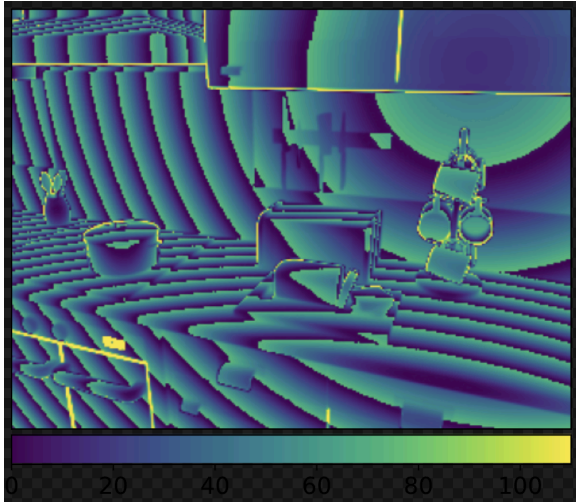


Figure GA1.a

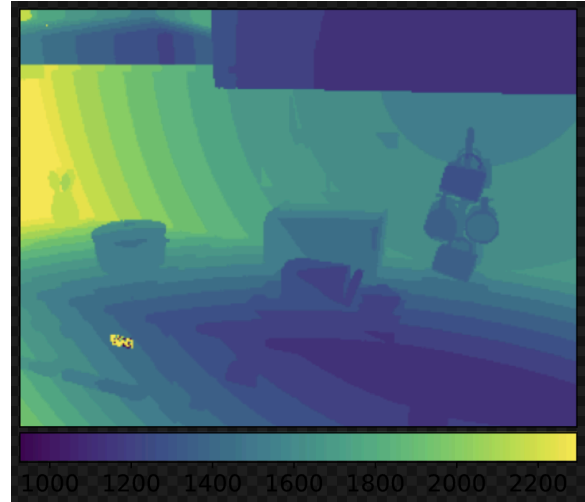


Figure GA1.b

7. **Identity Coding:** Identity coding, which corresponds to the full, uncompressed histogram, serves as the reference baseline for the Kitchen scene. The reconstructed depth map (Figure I1.b) is the smoothest and most accurate of all methods, with surfaces and edges that closely match the image. The absolute depth error map (Figure I1.a) contains the lowest error levels overall, with only small residual discrepancies in particularly challenging regions. As expected, the MAE of 9.58 mm for Identity coding was the lowest among all tested schemes. Notably, several compressed schemes, especially PSeries Gray and PSeries Fourier, achieved depth and depth error characteristics that come surprisingly close to this full-histogram baseline, reinforcing the central motivation behind CSPH.



Figure I1.a

Figure I1.b

4.2. Results on the Bathroom Scene

1. **Fourier Coding:** In the Bathroom scene, Fourier coding performed particularly poorly. The reconstructed depth map (Figure F2.b) is dominated by noise and random fluctuations, with very little recognizable scene structure. Surfaces that should appear smooth, such as walls and fixtures, instead show chaotic depth variations. The absolute depth error map (Figure F2.a) reflects this failure: error values are high across almost the entire image, with very few regions of acceptable accuracy. The MAE of 330.65 mm indicates severe degradation in depth estimation under the low-photon regime (100 photons). These results directly confirm the CSPH paper's observation that pure Fourier coding is highly unstable when photon counts are low and is generally unsuitable for robust compressive SPAD imaging.

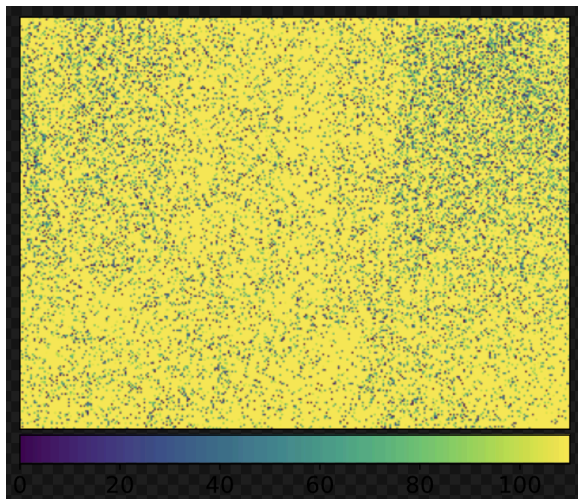


Figure F2.a

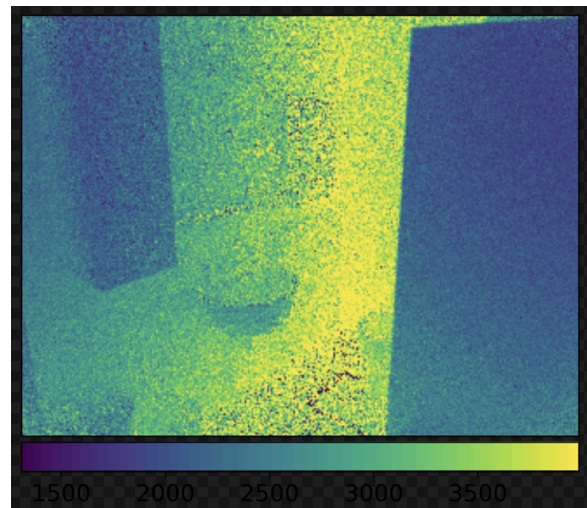
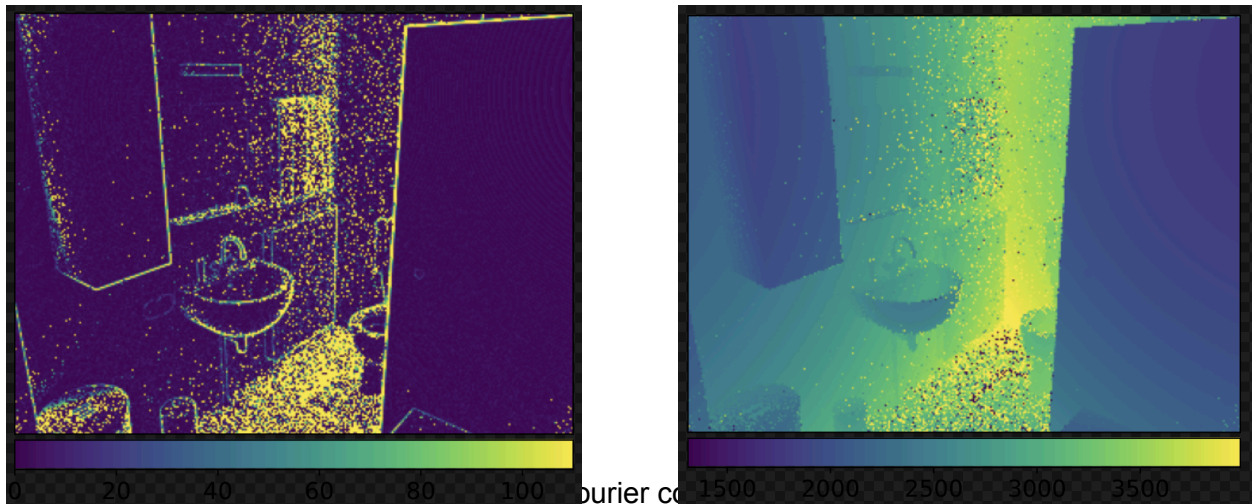


Figure F2.b

2. **Gray Coding:** Gray coding had a more balanced result in the Bathroom scene. The reconstructed depth map (Figure G2.b) recovers the general structure of the scene, and large surfaces appear roughly at the correct depth. However, due to the low photon

count and challenging lighting, certain regions exhibit significant depth failures. The absolute depth error map (Figure G2.a) reveals that much of the scene has moderate error, but scattered regions show huge error spikes. The MAE of 82.43 mm is elevated, but the median error remains relatively low, indicating that many pixels are still reconstructed accurately while a subset of outliers drives up the mean. This behavior is consistent with the CSPH paper's description of Gray coding, locality-preserving, and generally robust, yet susceptible to errors in particularly photon-starved or ambiguous regions.



best-performing compressed schemes in the Bathroom scene. The depth reconstruction (Figure PSF2.b) is noticeably more stable than pure Fourier or Truncated Fourier, with recognizable geometry and smoother surfaces despite the low photon budget. In the absolute depth error map (Figure PSF2.a), large regions exhibit moderate error, with fewer extreme outliers than in other methods. The resulting MAE of 51.81 mm is significantly lower than that of Fourier and Truncated Fourier and competitive with Gray approaches. While the reconstruction still lacks some fine detail due to photon scarcity, the overall depth and error characteristics strongly support the CSPH paper's claim that complementary frequency-based codes improve robustness under noisy, photon-limited conditions.

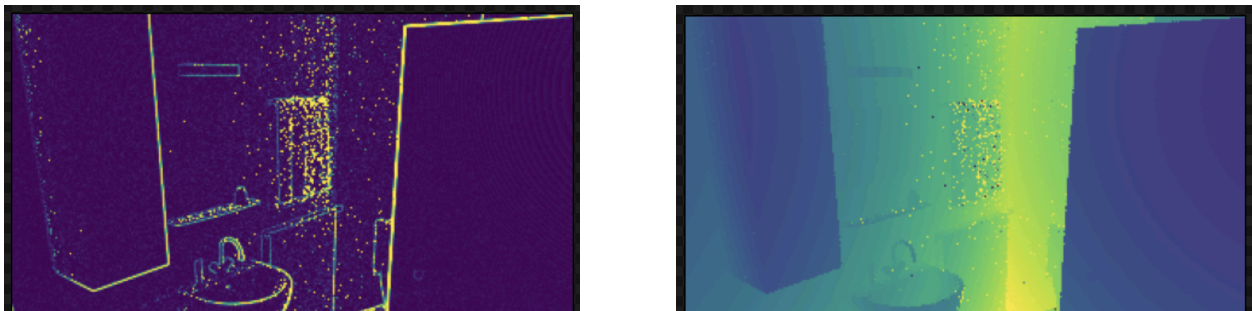


Figure PSF2.a

Figure PSF2.b

4. **Truncated Fourier Coding:** Truncated Fourier coding provided a partial improvement over pure Fourier in the Bathroom scene, but still fell short of PSeries-based methods. The reconstructed depth map (Figure TF2.b) shows a mix of smoother regions and noisy patches, with certain areas exhibiting pronounced spatial inconsistency. The absolute depth error map (Figure TF2.a) reveals a similar pattern: error levels are reduced compared to full Fourier but remain substantially higher and more irregular than those seen with Gray or PSeries Gray. The MAE of 65.69 mm occupies a mid-range value, reinforcing the idea that truncation can mitigate the sensitivity of Fourier coding to photon noise and multipath effects.

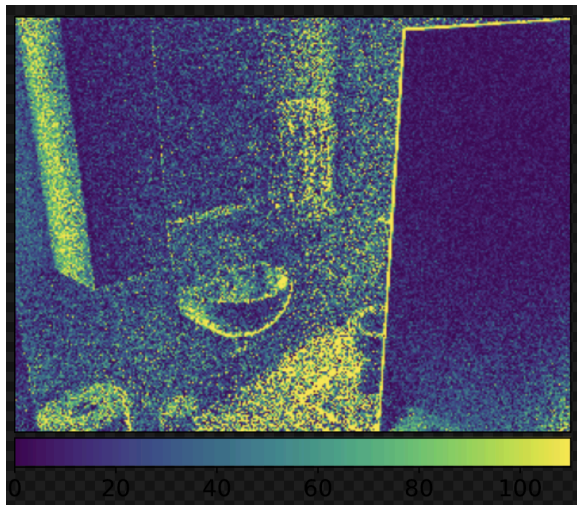


Figure TF2.a

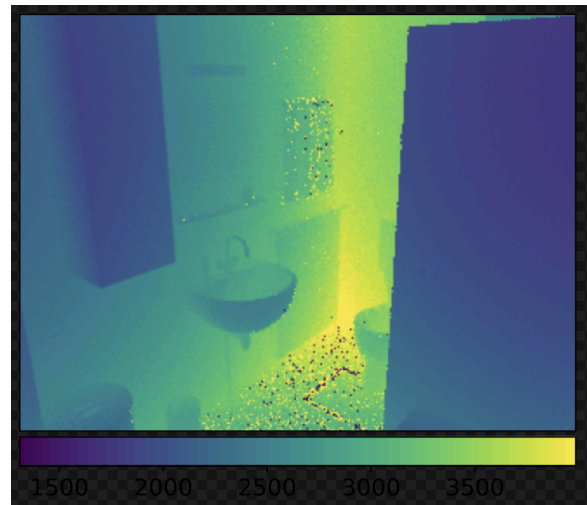
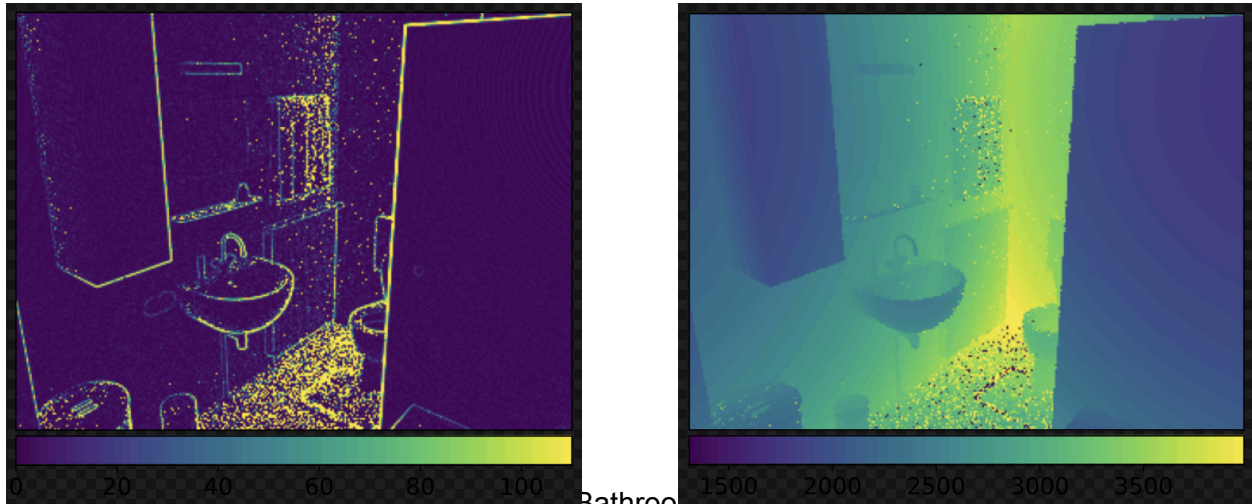


Figure TF2.b

5. **PSeries Gray Coding:** PSeries Gray coding achieved some of the strongest results among compressed schemes in the Bathroom experiment. The reconstructed depth map (Figure PSG2.b) preserves most large-scale structures and displays relatively smooth surfaces, even under the demanding low-photon conditions. The absolute depth error map (Figure PSG2.a) is comparatively clean. Many regions show low to moderate error, with fewer large outliers than with plain Gray or Fourier-based codes. As a result, the

MAE of 49.25 mm is among the lowest across all compressed strategies tested on this scene. This outcome provides further empirical support for the CSPH paper's central conclusion that Gray–Fourier designs, such as PSeries Gray, offer exceptional robustness and depth accuracy under compressive sampling.



6. **Gated Coding.** Gated coding in the Bathroom scene behaves similarly to its Kitchen counterpart, but with additional degradation due to lower photon counts. The reconstructed depth map (Figure GA2.b) appears with banded depth levels and limited ability to represent subtle gradients or small depth variations. The absolute depth error map (Figure GA2.a) shows widespread moderate error and frequent artifacts near edges and low-return regions. The MAE of 63.02 mm is higher than that of PSeries schemes, reflecting the method's limited effective temporal resolution. Under these challenging conditions, gating provides only a coarse approximation of scene depth and does not reach the accuracy levels required for high-quality 3D imaging.

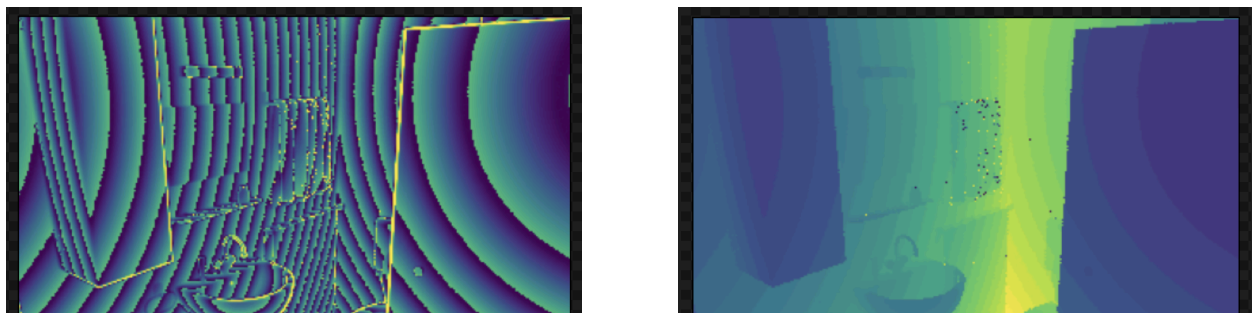


Figure GA2.a

Figure GA2.b

7. **Identity Coding:** Identity coding in the Bathroom scene is in contrast to the Kitchen results. While the reconstructed depth map (Figure I2.b) captures the overall structure reasonably well, certain regions suffer from large errors due to photon statistics and a strong background. The absolute depth error map (Figure I2.a) shows many pixels with low error, but also a noticeable set of high-error outliers. The median depth error remains very low, but the MAE of 75.92 mm is surprisingly high because of these extreme failures. This behavior illustrates a key insight emphasized in the CSPH paper. Full histograms are not automatically optimal in the low-photon regime, and appropriately designed compressed coding schemes can outperform them by aggregating the signal more effectively. In the Bathroom scene, several compressed strategies achieve more balanced depth and error profiles than the full-histogram baseline, underscoring the practical advantages of CSPH under realistic photon-limited conditions.

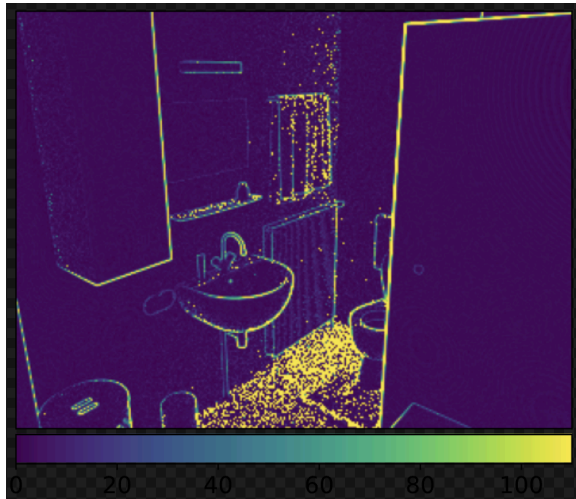


Figure I2.a

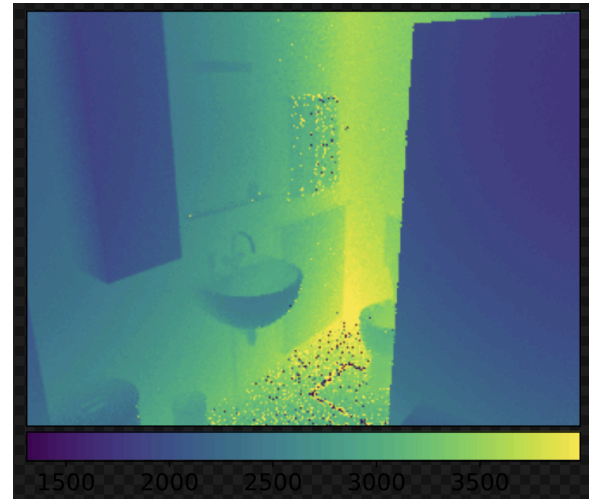


Figure I2.b

4.3. Summary of Findings Across Both Scenes

Across both the Kitchen and Bathroom simulations, several consistent performance patterns emerged. Fourier coding consistently showed the poorest stability, producing noisy

depth maps and high MAE, especially in photon-limited scenarios, confirming the CSPH paper's conclusion that frequency-only codes lack locality and robustness. Gray coding performed much better, generating smooth depth reconstructions with relatively low error, though occasional ambiguous bins led to large outliers in the low-photon Bathroom scene. Gated coding remained coarse and limited in precision, with visibly quantized depth outputs and structured depth error patterns that reflected its stepwise time-window design.

The P-Series variants consistently produced strong reconstructions in these experiments. In practice, their depth maps appeared more stable and their depth error maps cleaner, often yielding lower MAE than the corresponding base scheme in these runs. The Identity (full histogram) coding behaved as expected in the Kitchen scene, producing the lowest error and serving as an upper-bound reference. However, in the Bathroom experiment, sparse photon returns caused extreme outliers that raised the MAE unexpectedly high, illustrating that full histograms are not inherently optimal in low-signal conditions. Notably, several compressed coding strategies outperformed or matched the full histogram in these challenging settings. These findings reinforce the central claims of the CSPH framework: well-designed compressive codes can deliver high-quality depth reconstructions while dramatically reducing data bandwidth. Overall, the simulations validate the theoretical advantages of compressed coding for modern SPAD-based 3D imaging systems.

5. Pulse Width Analysis on Truncated Fourier Coding

To further understand how different coding parameters influence reconstruction quality, I changed the pulse width on the Kitchen scene using Truncated Fourier coding. I tested three different pulse widths, $PW = 1, 16, \text{ and } 64$, representing narrow, moderate, and wide spreads, respectively. This test was done to see how pulse widths affect both depth reconstruction and the absolute depth error distribution, offering insight into how coding design interacts with the physical characteristics of a SPAD-based time-of-flight system.

1. **Pulse Width = 1:** The system produces a very sharp, almost ideal impulse response. This gives high temporal precision, but it also makes Truncated Fourier coding extremely sensitive to noise. As a result, the reconstructed depth map (Figure PW1.b) shows a lot of fluctuations and instability, especially on flat surfaces. The depth error map (Figure PW1.a) confirms this, with large errors spread across much of the scene. The MAE of 22.37 mm remains high because the coding scheme cannot handle this level of precision without stronger locality, a limitation also noted in the CSPH paper.

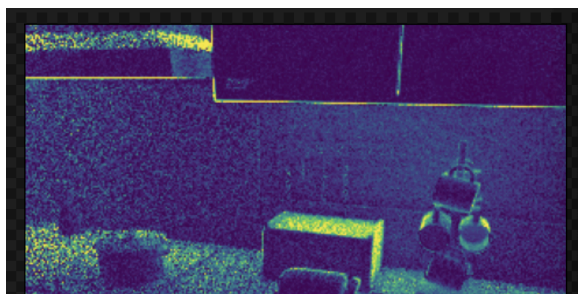


Figure PW1.a

Figure PW1.b

2. **Pulse Width = 16:** The reconstruction quality improves significantly. The slightly wider impulse response smooths out high-frequency noise, reducing the oscillations seen with $PW = 1$. The depth map (Figure PW16.b) becomes more consistent across surfaces, and the depth error map (Figure PW16.a) shows noticeably lower error. This leads to a lower MAE of 23.32 mm overall. This level of smoothing helps Truncated Fourier coding work more reliably because it reduces instability while still preserving important depth information.

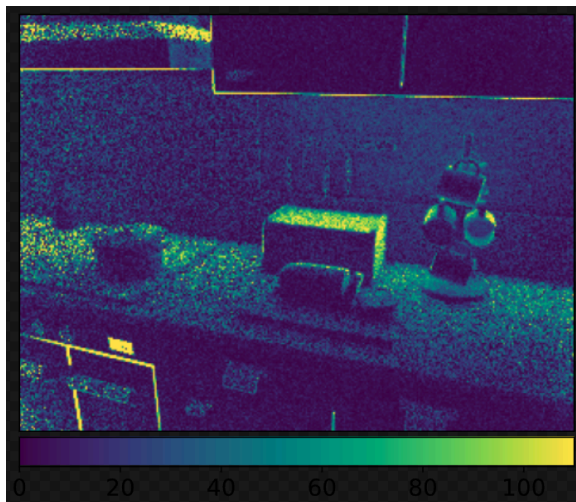


Figure PW16.a

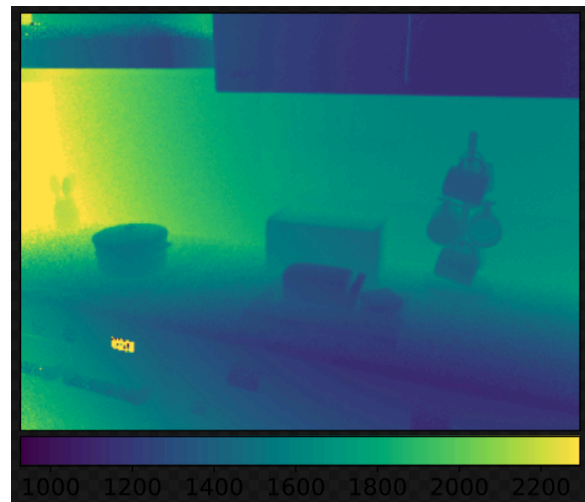
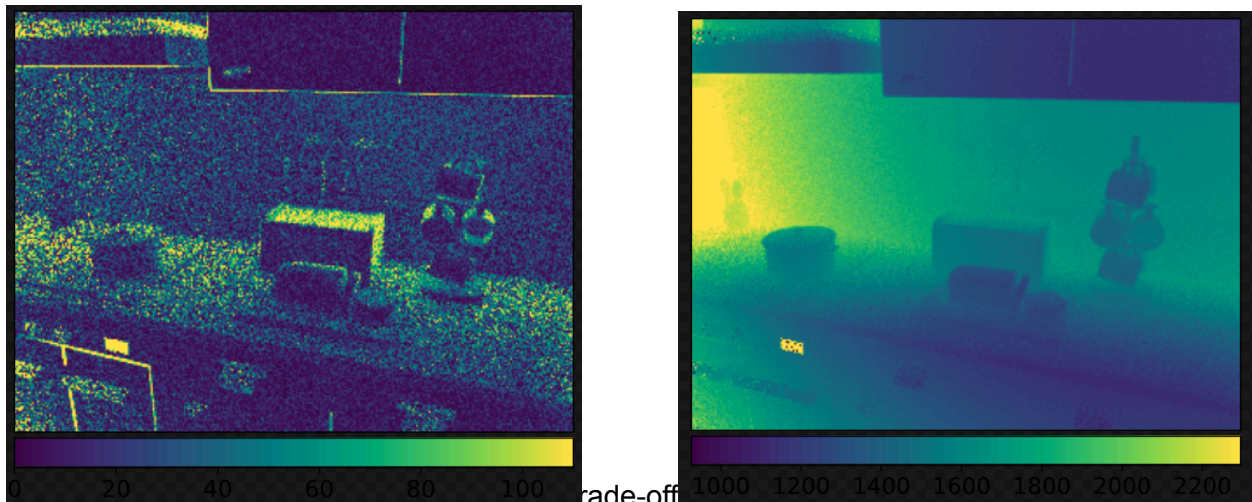


Figure PW16.b

3. **Pulse Width = 64:** The impulse response becomes very wide, heavily smoothing the signal. This produces the cleanest and most stable depth reconstruction (Figure PW64.b) of the three settings, with the lowest MAE of 30.44 mm and very low error in most of the depth error map (Figure PW64.a). However, this comes at a cost, where fine details and sharp depth transitions begin to blur. As the CSPH paper explains, overly wide pulse widths reduce the system's effective temporal resolution. So while $PW = 64$ is stable, it sacrifices detail, demonstrating the trade-off between smoothing and depth precision.



width (1) provides high temporal resolution but results in unstable, noise-sensitive depth estimates. A moderate pulse width (16) offers the best balance, significantly reducing MAE while preserving important scene details. A large pulse width (64) produces the most stable and lowest-error reconstructions, though at the cost of blurring fine depth features. Overall, these findings reinforce a key point from the CSPH paper. Coding performance is closely tied to the system's impulse response, and for Truncated Fourier coding, moderate to wide pulse widths help improve stability by better matching the temporal behavior of the system.

6. Conclusion

Through this replication study, several clear trends emerged in the performance of compressive coding schemes for SPAD-based depth imaging. Across both the Kitchen and Bathroom scenes, hybrid approaches, especially PSeries Gray and PSeries Fourier, consistently produced the most accurate and stable reconstructions, often nearing the performance of the full histogram while operating at drastically reduced data bandwidth. Truncated Fourier struggled with noise sensitivity, particularly in low-photon environments. These experiments validate the core findings of the CSPH paper, with well-designed coding matrices, and high-quality depth estimation is possible even under aggressive compression.

The pulse-width experiments further showed that parameters play a crucial role in reconstruction quality. Narrow impulse responses (pulse width 1) amplified instability in Truncated Fourier coding, while moderate to wide pulse widths stabilized reconstructions and reduced MAE. However, excessively wide pulse widths also blurred fine details, revealing a trade-off between temporal resolution and robustness. These results confirm that coding design and sensor behavior must be jointly optimized to achieve reliable depth imaging.

While the results demonstrate the power of CSPH and modern coding strategies, they also highlight important limitations that shape what remains to be solved. Compressive coding requires significantly more computation than traditional histograms, since each photon

contributes to multiple coded bins. This overhead could be mitigated with future hardware acceleration, but it currently poses a challenge for real-time, high-resolution SPAD systems.

The replication results reinforce the main message of the CSPH framework that compressive coding offers a promising, scalable path forward, enabling SPAD systems to achieve high-precision depth imaging while dramatically reducing data rates. Yet realizing the full potential of these methods will require continued progress in hardware, algorithmic innovation, and thoughtful integration into real-world systems.